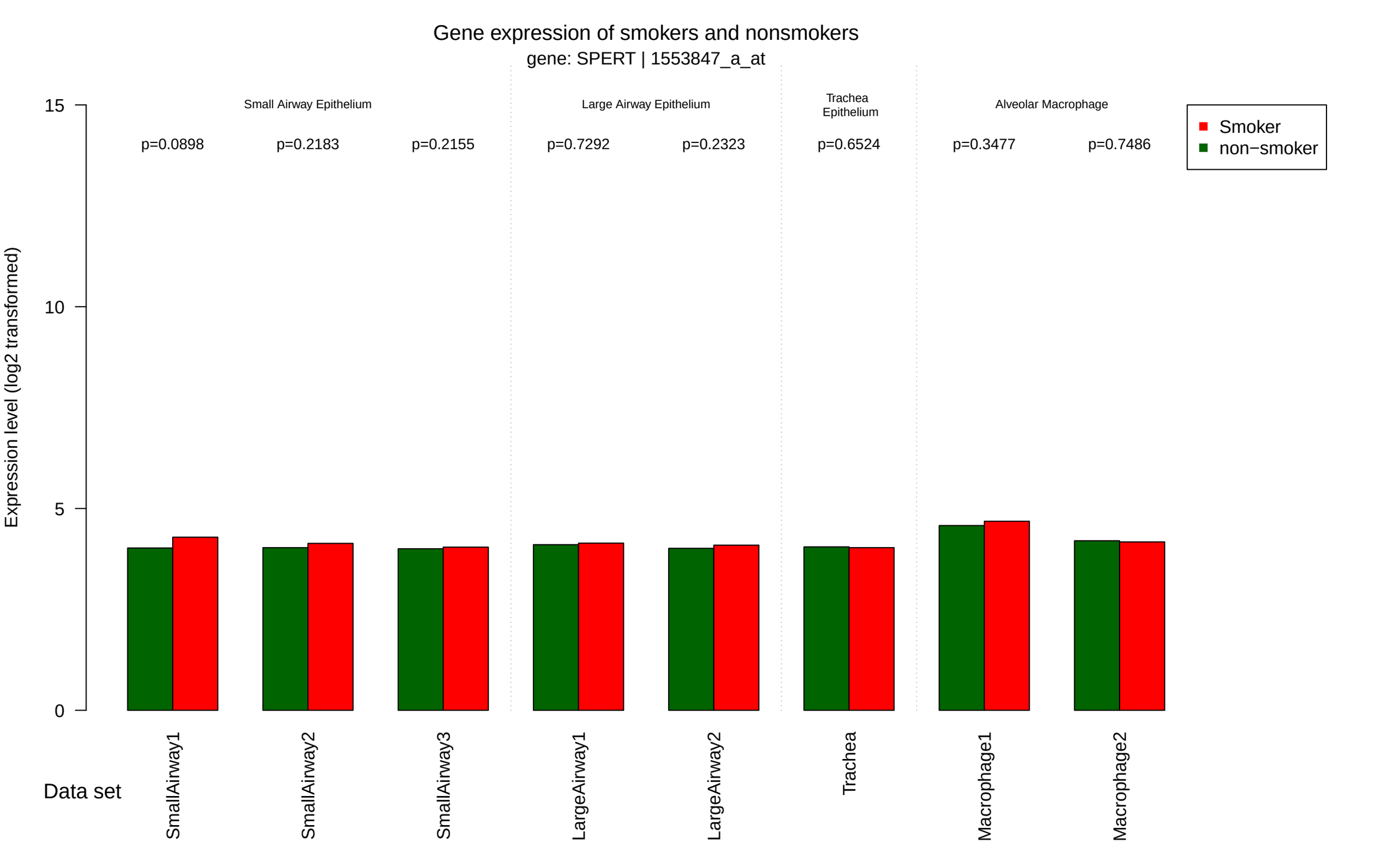
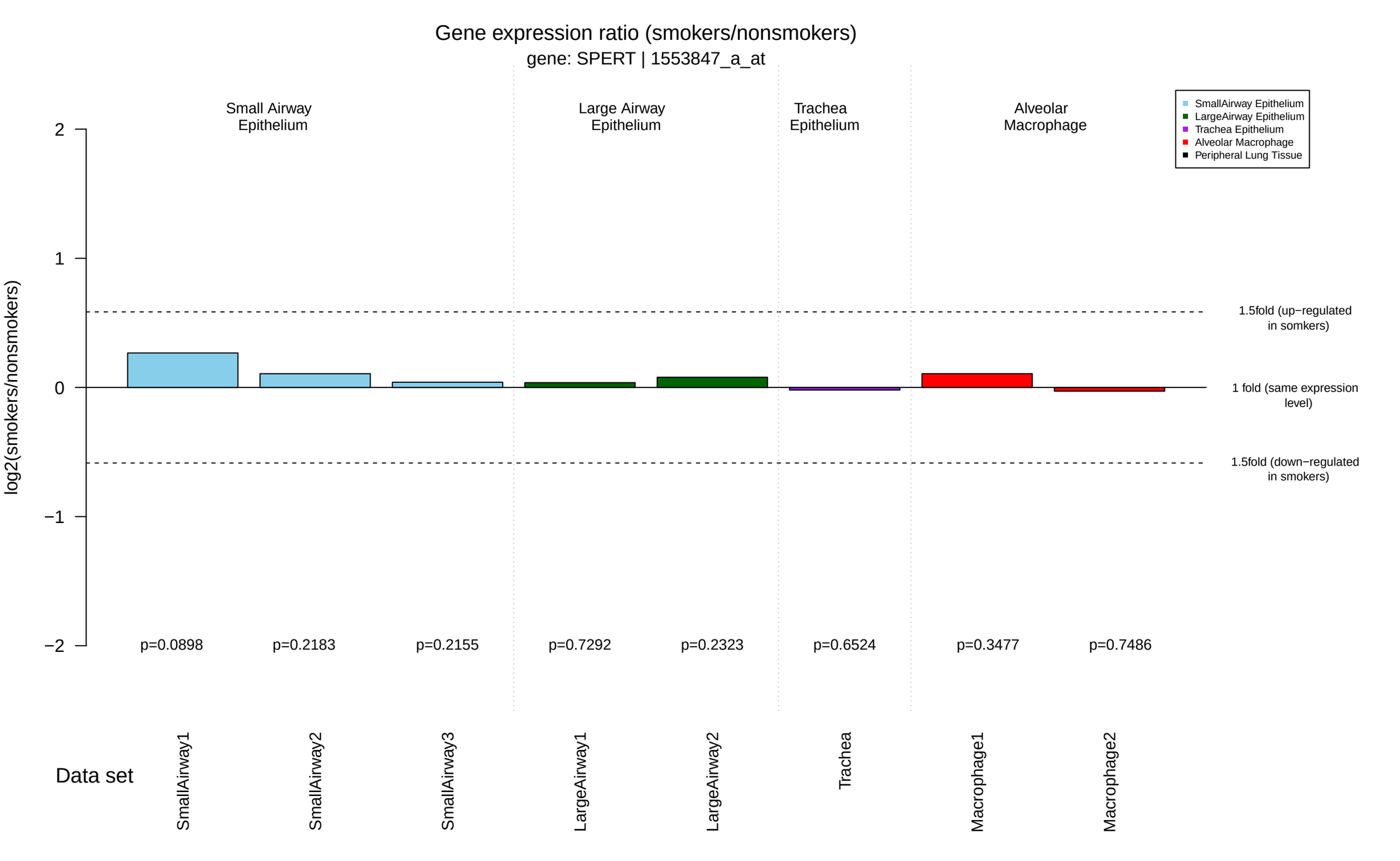
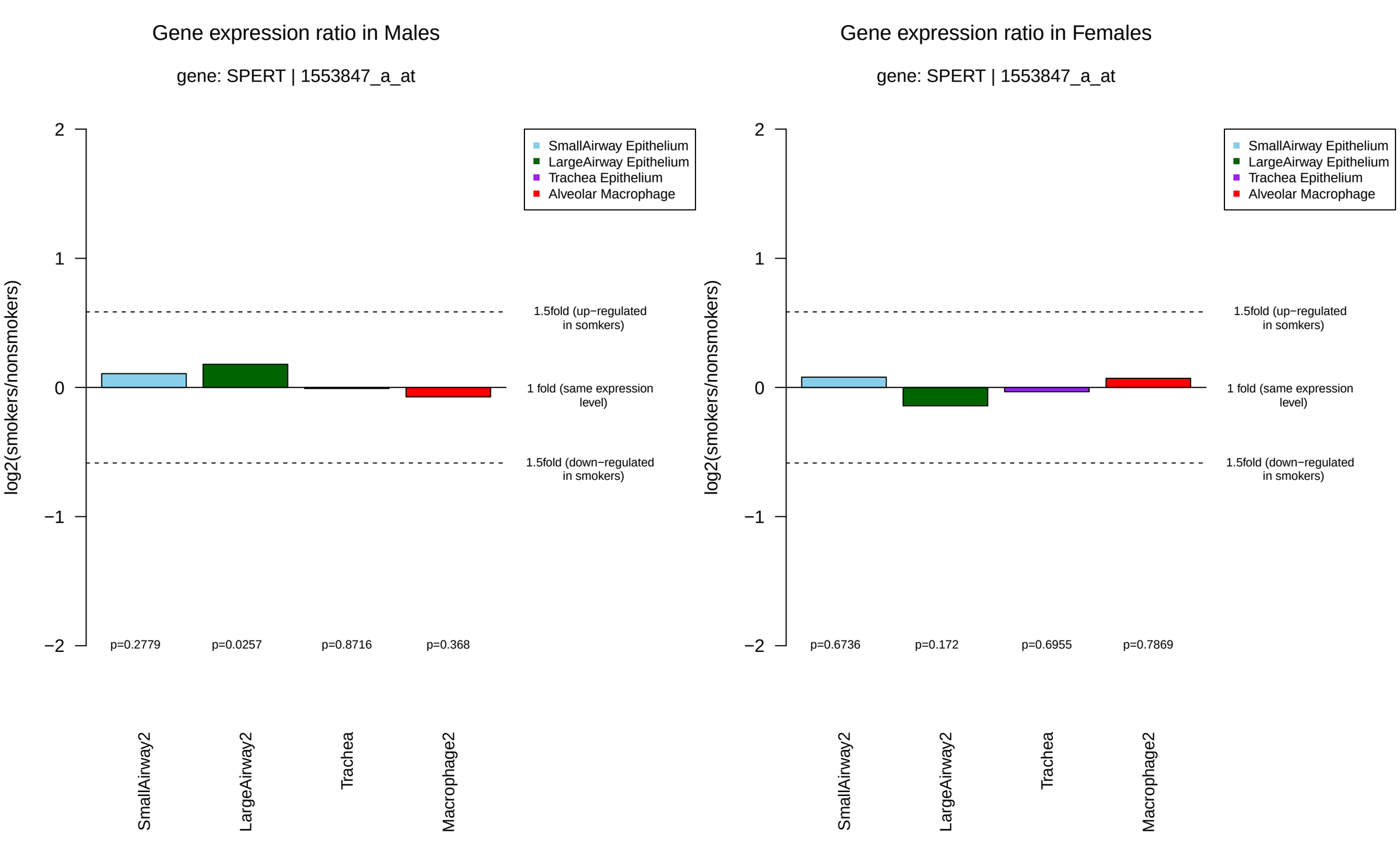
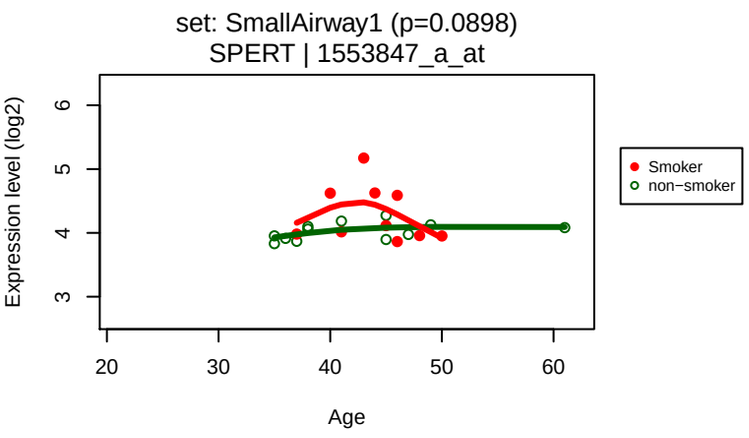
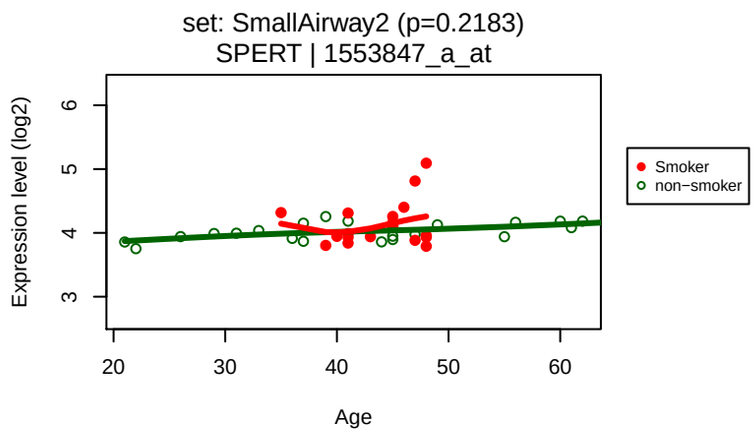
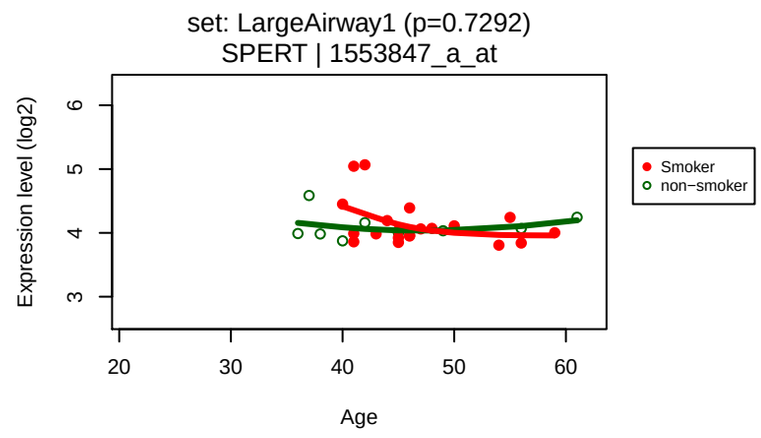

Gene expression of smokers and nonsmokers
gene: SPERT | 1553847_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0898
p=0.2183
p=0.2155
p=0.7292
p=0.2323
p=0.6524
p=0.3477
p=0.7486
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

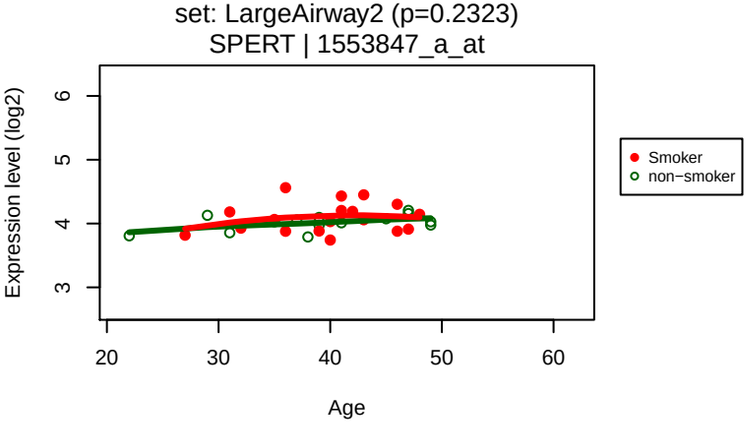

Gene expression ratio (smokers/nonsmokers)
gene: SPERT | 1553847_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0898
p=0.2183
p=0.2155
p=0.7292
p=0.2323
p=0.6524
p=0.3477
p=0.7486
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

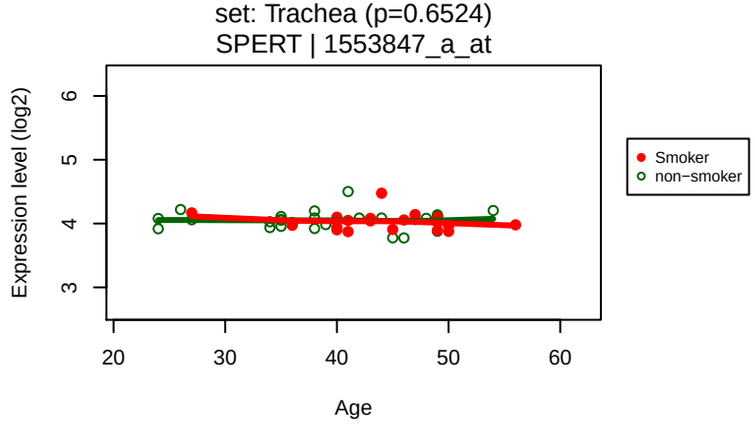

Gene expression ratio in Males
gene: SPERT | 1553847_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2779
p=0.0257
p=0.8716
p=0.368
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPERT | 1553847_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6736
p=0.172
p=0.6955
p=0.7869
SmallAirway2
LargeAirway2
Trachea
Macrophage2

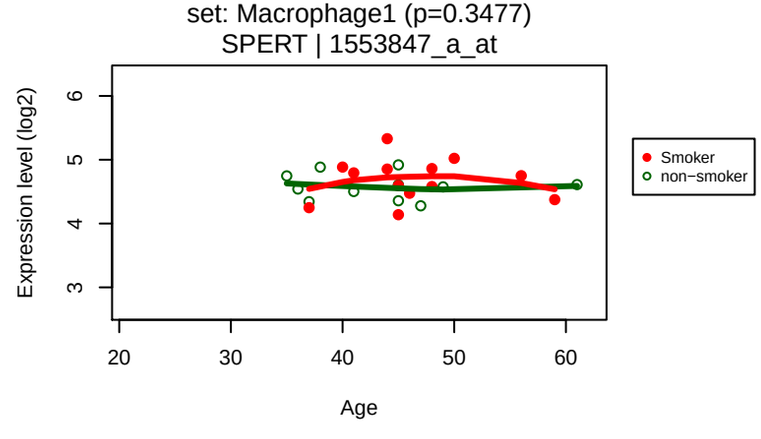

set: SmallAirway1 (p=0.0898)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker

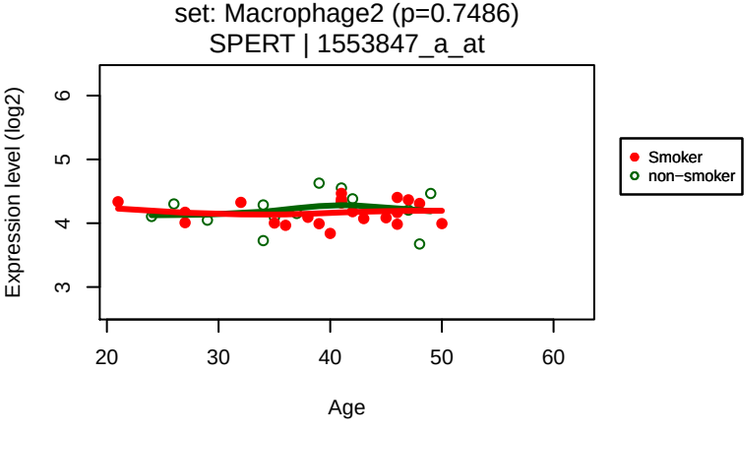

set: SmallAirway2 (p=0.2183)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker

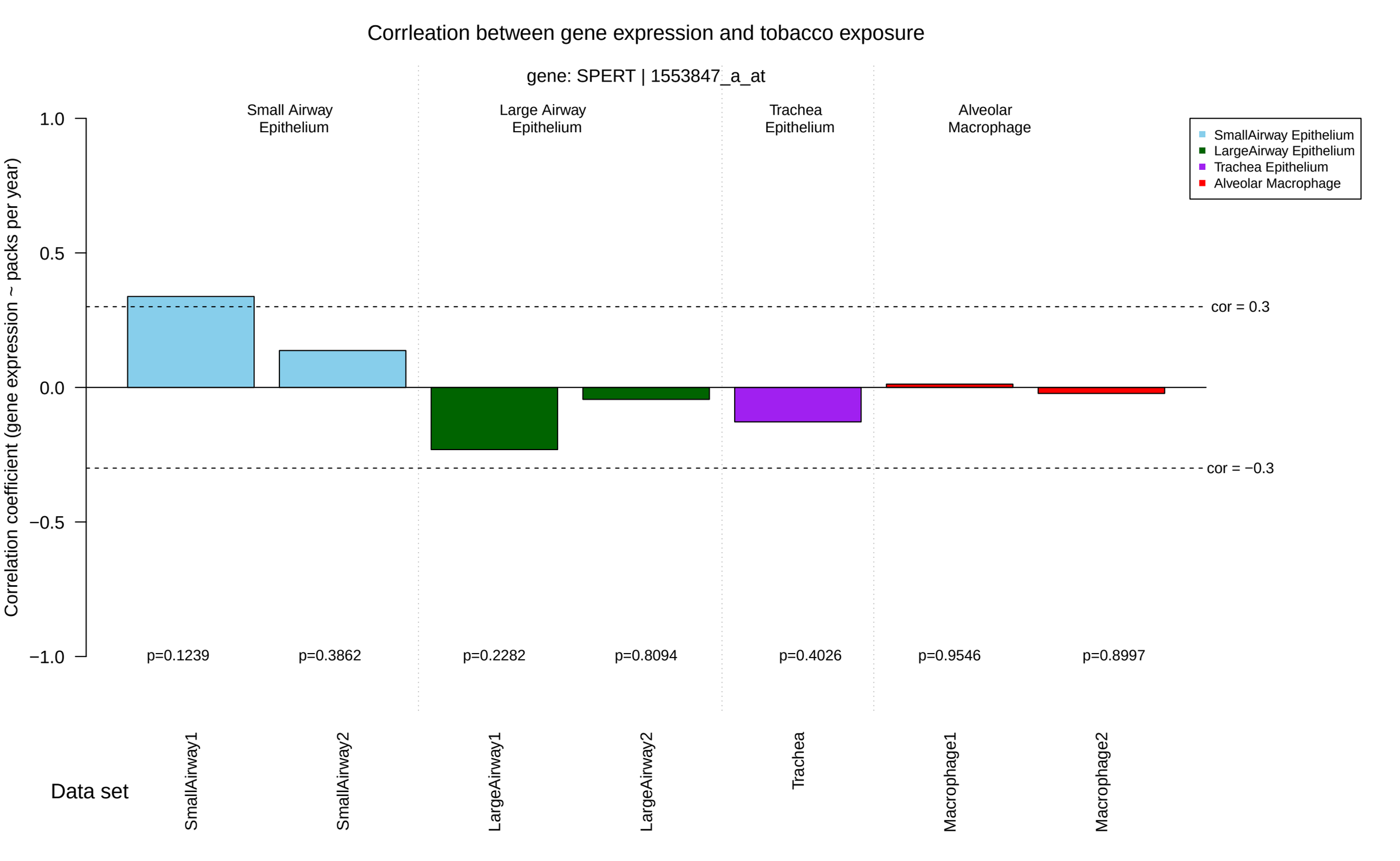

set: LargeAirway1 (p=0.7292)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2323)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6524)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3477)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7486)
SPERT | 1553847_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPERT | 1553847_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1239
p=0.3862
p=0.2282
p=0.8094
p=0.4026
p=0.9546
p=0.8997
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2